

Climate 5000 Series - Gen 4

36K BTU Single Zone Light Commercial System

Wall Mounted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

36K Btu Single Zone Light Commercial System - Wall Mounted IDU

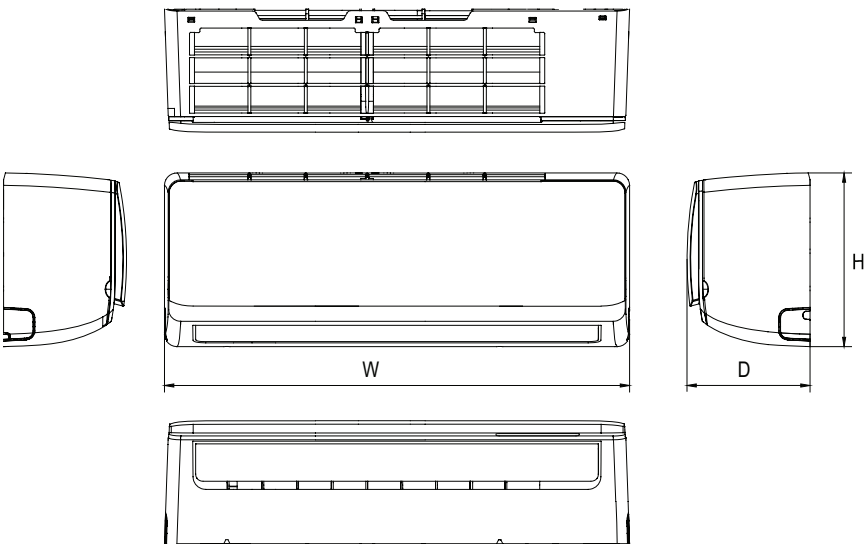
Indoor part number		8-733-962-679
Indoor model number		BMS500-AAU036-1AHWXD
Outdoor part number		8-733-962-680
Outdoor model number		BMS500-AAS036-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		19.0
EER2		8.6
HSPF2 Region IV		8.5
HSPF2 Region V		6.6
COP(47F)		2.85
COP (5F)		1.90
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	6600~36800
	Rated Capacity	36000
	Current	18.0
	Input Power	4138
Heating	Capacity Range	9600~38000
	Rated Capacity (47F)	36000
	Rated Capacity (5F)	22000
	Current	16.2
	Input Power	3700
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 33.0
Max. fuse indoor/outdoor (A)		15.0 / 35.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 670
Input (W)		2200
Rated current (RLA/A)		10.0
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

Piping		
Refrigerant precharge (oz/kg)	112.88 / 3.2	
Length for precharge (ft/m)	24.6 / 7.5	
Liquid side (in/mm)	3/8" / Ø9.52	
Gas side (in/mm)	5/8" / Ø15.9	
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30	
Max. refrigerant pipe length (ft/m)	213.20 / 65	
Max. difference in height (ft/m)	98.40 / 30	
System Data		
Refrigerant type	R454B	
Design pressure (PSIG)	550 / 340	
Dehumidification (pts/hr)	9.72	
IDU	Fan motor model	ZKFP-58-8-1-10
	Fan motor qty (ea)	1
	Fan motor input (W)	58.0
	Fan speed (RPM)	1050 / 880 / 630
	Air flow, Turbo / H / M / L / Si (CFM)	882.9 / 635.7 / 506.2 / 382.6 / 323.7
	Approx. Air Throw Range ft (m)	29.5 / 9
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 49 / 43.5 / 34.5 / 31.5
	Net weight (lbs / kg)	43.21 / 19.6
	Gross weight (lbs / kg)	54.67 / 24.8
ODU	Fan motor model	ZKFN-120-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	170
	Fan motor speed, H / M / L (RPM)	880 / 700 / 500
	Air flow, Max (CFM)	2235.3
	Noise level, Max (dB(A))	63.0
	Net weight (lbs / kg)	150.35 / 68.2
	Gross weight (lbs / kg)	160.27 / 72.7
Accessories		
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat	
BMS-DRY-XXD (8733969263)	24V Interface	
BMS-PK3-7X1 (8733951015)	25ft Lineset	
Standard Parts		
8-733-969-088	Remote Control	

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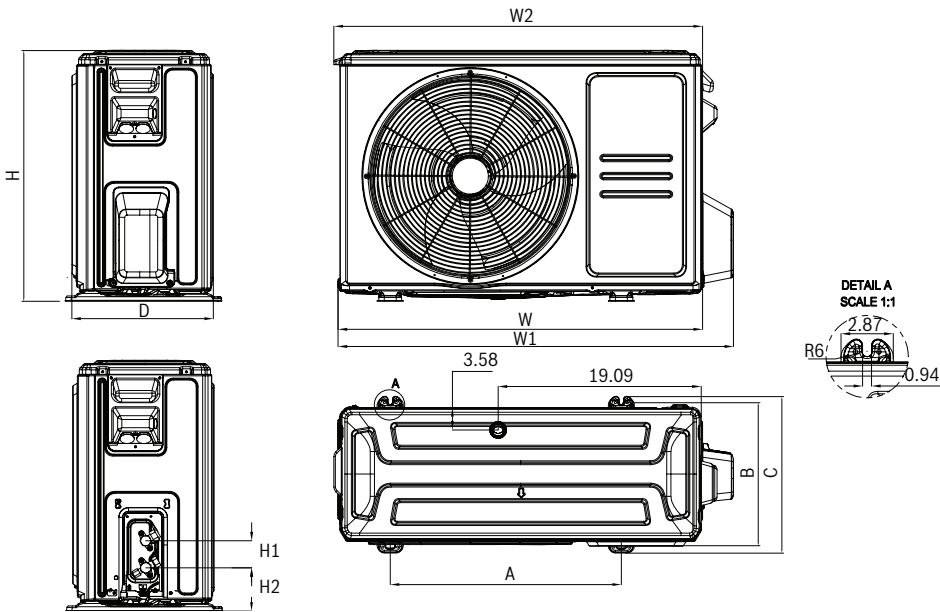


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU036-1AHWXD	44.88	11.22	14.61

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS036-1CSXRD	37.24	16.14	31.89	40.55	37.24	2.36	4.25	26.5	15.87	17.9

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Heating Performance - Indoor Dry Bulb Temperature at 882 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	27016	27053	27082	27083	27120	28645	28679	28707	28713	28747	30356	30395	30429	30435	30469	32185	32220	32255	32260	32300
	S/T	0.48	0.56	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.77	0.54	0.63	0.72	0.72	0.82	0.57	0.67	0.76	0.76	0.87
	PI	1.60	1.60	1.60	1.60	1.60	1.69	1.69	1.70	1.70	1.70	1.79	1.80	1.80	1.80	1.80	1.90	1.90	1.91	1.91	1.91
-4°F (-20.0°C)	TC	29095	29134	29165	29166	29206	30848	30885	30915	30921	30958	32691	32734	32770	32776	32813	34661	34698	34736	34742	34785
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.82	0.57	0.67	0.77	0.77	0.87
	PI	1.72	1.72	1.72	1.72	1.73	1.82	1.83	1.83	1.83	1.83	1.93	1.93	1.94	1.94	1.94	2.05	2.05	2.05	2.05	2.06
0°F (-17.8°C)	TC	30018	30059	30091	30092	30133	31827	31865	31897	31903	31941	33729	33773	33810	33817	33854	35761	35800	35839	35845	35889
	S/T	0.48	0.57	0.64	0.64	0.73	0.51	0.60	0.68	0.68	0.78	0.54	0.64	0.72	0.72	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.77	1.78	1.78	1.78	1.78	1.88	1.88	1.89	1.89	1.89	1.99	2.00	2.00	2.00	2.00	2.11	2.12	2.12	2.12	2.12
5°F (-15.0°C)	TC	31173	31215	31248	31250	31292	33051	33091	33123	33130	33169	35026	35072	35111	35117	35156	37137	37177	37217	37223	37270
	S/T	0.49	0.57	0.65	0.65	0.74	0.51	0.60	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.84	1.84	1.85	1.85	1.85	1.95	1.96	1.96	1.96	1.96	2.07	2.07	2.08	2.08	2.08	2.19	2.20	2.20	2.20	2.20
14°F (-10.0°C)	TC	32094	32137	32172	32173	32216	34028	34068	34102	34109	34149	36061	36108	36148	36155	36195	38234	38275	38317	38323	38371
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.78	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88
	PI	1.75	1.75	1.76	1.76	1.76	1.86	1.86	1.86	1.86	1.86	1.97	1.97	1.97	1.97	1.97	2.09	2.09	2.09	2.09	2.09
17°F (-8.3°C)	TC	32401	32444	32479	32481	32525	34353	34394	34428	34435	34476	36406	36453	36494	36501	36541	38599	38641	38683	38689	38738
	S/T	0.49	0.57	0.65	0.65	0.74	0.52	0.61	0.69	0.69	0.79	0.55	0.64	0.73	0.73	0.83	0.58	0.68	0.78	0.78	0.88
	PI	1.72	1.72	1.72	1.72	1.73	1.82	1.83	1.83	1.83	1.83	1.93	1.94	1.94	1.94	1.94	2.05	2.05	2.05	2.05	2.06
23°F (-5.0°C)	TC	32435	32479	32514	32515	32559	34390	34430	34464	34471	34512	36444	36492	36532	36539	36580	38640	38682	38724	38730	38778
	S/T	0.49	0.58	0.65	0.65	0.75	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.58	0.69	0.78	0.78	0.89
	PI	1.74	1.74	1.74	1.74	1.75	1.84	1.85	1.85	1.85	1.85	1.95	1.96	1.96	1.96	1.96	2.07	2.07	2.08	2.08	2.08
35°F (1.7°C)	TC	32503	32547	32582	32583	32627	34462	34503	34537	34544	34585	36521	36568	36609	36616	36657	38721	38763	38805	38812	38860
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.77	1.78	1.78	1.78	1.78	1.88	1.88	1.88	1.89	1.89	1.99	2.00	2.00	2.00	2.00	2.11	2.12	2.12	2.12	2.12
41°F (5.0°C)	TC	34291	34337	34374	34375	34422	36357	36400	36436	36444	36487	38529	38579	38622	38630	38673	40851	40895	40940	40946	40997
	S/T	0.49	0.58	0.66	0.66	0.75	0.52	0.61	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89
	PI	1.82	1.82	1.83	1.83	1.83	1.93	1.93	1.93	1.93	1.94	2.05	2.05	2.05	2.05	2.05	2.17	2.17	2.17	2.17	2.18
47°F (8.3°C)	TC	36078	36127	36166	36168	36216	38253	38298	38336	38343	38389	40538	40591	40636	40644	40689	42981	43027	43074	43081	43135
	S/T	0.50	0.58	0.66	0.66	0.75	0.53	0.62	0.70	0.70	0.80	0.56	0.65	0.74	0.74	0.85	0.59	0.69	0.79	0.79	0.90
	PI	1.87	1.87	1.87	1.87	1.88	1.98	1.98	1.98	1.99	1.99	2.10	2.10	2.10	2.10	2.11	2.23	2.23	2.23	2.23	2.23
50°F (10.0°C)	TC	36123	36172	36211	36213	36261	38300	38346	38384	38391	38437	40589	40641	40687	40694	40740	43034	43081	43128	43135	43188
	S/T	0.50	0.58	0.66	0.66	0.76	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90
	PI	1.91	1.92	1.92	1.92	1.92	2.03	2.03	2.03	2.03	2.04	2.15	2.15	2.16	2.16	2.16	2.28	2.28	2.29	2.29	2.29
59°F (15.0°C)	TC	36258	36307	36346	36348	36397	38443	38489	38527	38535	38580	40740	40793	40839	40846	40892	43195	43242	43289	43296	43350
	S/T	0.50	0.59	0.67	0.67	0.76	0.53	0.62	0.71	0.71	0.81	0.56	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91
	PI	2.05	2.05	2.06	2.06	2.06	2.18	2.18	2.18	2.18	2.18	2.31	2.31	2.31	2.31	2.31	2.44	2.45	2.45	2.45	2.45
68°F (20.0°C)	TC	36393	36442	36481	36483	36532	38586	38632	38670	38678	38724	40892	40945	40990	40998	41044	43356	43403	43450	43457	43511
	S/T	0.51	0.60	0.69	0.69	0.78	0.54	0.64	0.73	0.73	0.83	0.58	0.68	0.77	0.77	0.88	0.61	0.72	0.82	0.82	0.93
	PI	2.19	2.19	2.20	2.20	2.20	2.32	2.32	2.33	2.33	2.33	2.46	2.46	2.47	2.47	2.47	2.61	2.61	2.61	2.61	2.62
77°F (25.0°C)	TC	36528	36577	36617	36618	36668	38729	38775	38814	38821	38867	41043	41097	41142	41150	41196	43516	43564	43611	43618	43672
	S/T	0.52	0.61	0.69	0.69	0.79	0.55	0.65	0.74	0.74	0.84	0.59	0.69	0.78	0.78	0.89	0.62	0.73	0.83	0.83	0.94
	PI	2.33	2.33	2.33	2.33	2.34	2.47	2.47	2.47	2.47	2.48	2.62	2.62	2.62	2.62	2.63	2.77	2.78	2.78	2.78	2.78
82°F (27.8°C)	TC	36059	36108	36147	36148	36197	38232	38278	38315	38323	38368	40516	40569	40614	40622	40667	42958	43004	43051	43058	43111
	S/T	0.53	0.62	0.70	0.70	0.80	0.56	0.66	0.75	0.75	0.85	0.59	0.70	0.79	0.79	0.90	0.63	0.74	0.84	0.84	0.95
	PI	2.57	2.58	2.58	2.58	2.58	2.73	2.73	2.73	2.73	2.74	2.89	2.89	2.90	2.90	2.90	3.06	3.07	3.07	3.07	3.08
86°F (30.0°C)	TC	34998	35045	35083	35084	35132	37107	37151	37188	37195	37239	39324	39375	39419	39426	39470	41693	41739	41784	41790	41843
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	2.98	2.98	2.99	2.99	2.99	3.16	3.16	3.17	3.17	3.17	3.35	3.35	3.36	3.36	3.36	3.55	3.55	3.56	3.56	3.56
95°F (35.0°C)	TC	32610	32654	32689	32690	32734	34575	34616	34650	34657	34698	36641	36688	36729	36736	36777	38848	38891	38933	38939	38988
	S/T	0.53	0.63	0.71	0.71	0.81	0.57	0.66	0.75	0.75	0.86	0.60	0.70	0.80	0.80	0.91	0.64	0.75	0.85	0.85	0.97
	PI	3.90	3.90	3.91	3.91	3.91	4.13	4.14	4.14	4.14	4.15	4.38	4.38	4.39	4.39	4.39	4.64	4.65	4.65	4.65	4.66
104°F (40.0°C)	TC	27209	27245	27274	27276	27312	28848	28882	28911	28917	28951	30572	30611	30646	3063						

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Heating Performance - Indoor Dry Bulb Temperature at 882 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	23800	23324	23208	23115	22907	0.00	0.00	0.00	0.00	0.00
-4°F(-20.0°C)	23547	23076	22961	22869	22663	0.00	0.00	0.00	0.00	0.00
0°F(-17.8°C)	23434	22966	22851	22760	22555	1.46	1.47	1.47	1.47	1.48
5°F(-15.0°C)	23294	22828	22714	22623	22419	3.28	3.30	3.31	3.32	3.33
14°F(-10.0°C)	23040	22580	22467	22377	22175	3.30	3.32	3.33	3.34	3.35
17°F(-8.3°C)	22956	22497	22384	22295	22094	3.30	3.33	3.34	3.34	3.36
22°F(-5.6°C)	23197	22733	22619	22529	22326	3.37	3.40	3.41	3.41	3.43
27°F(-2.8°C)	23438	22969	22854	22763	22558	3.44	3.47	3.47	3.48	3.50
32°F(0.0°C)	23679	23206	23090	22997	22790	3.51	3.54	3.54	3.55	3.56
35°F(1.7°C)	23824	23347	23231	23138	22929	3.55	3.58	3.58	3.59	3.61
37°F(2.8°C)	25763	25248	25121	25021	24796	2.96	2.98	2.99	2.99	3.00
42°F(5.6°C)	30611	29999	29849	29729	29462	1.48	1.49	1.49	1.50	1.50
44.6°F(7.0°C)	32550	31899	31740	31613	31328	0.89	0.89	0.90	0.90	0.90
47°F(8.3°C)	35459	34750	34576	34438	34128	0.00	0.00	0.00	0.00	0.00
52°F(11.1°C)	34588	33896	33727	33592	33289	0.56	0.56	0.56	0.57	0.57
57°F(13.9°C)	33717	33042	32877	32746	32451	1.12	1.13	1.13	1.13	1.14
59°F(15.0°C)	33368	32701	32538	32407	32116	1.34	1.35	1.35	1.36	1.36
62°F(16.7°C)	32846	32189	32028	31900	31613	1.68	1.69	1.69	1.70	1.70
64.4°F(18.0°C)	32497	31847	31688	31561	31277	1.90	1.92	1.92	1.92	1.93

Bolded values were tested. All other values are simulated and for reference only.